

FULL-LENGTH ORIGINAL RESEARCH

Seizure outcomes following multilobar epilepsy surgery

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SUMMARY

Purpose: Outcomes following unilobar surgeries for refractory epilepsy have been well described. However, little is known about long-term seizure outcomes following multilobar resections. The aim of the current study was to identify long-term seizure control and predictors of seizure recurrence in this patient population.

Methods: Records of patients who underwent multilobar epilepsy surgery at the Cleveland Clinic between 1994 and 2010 were retrospectively reviewed. A postoperative follow-up of at least 6 months was required. Patients were classified as seizure free if they achieved an Engel class I at last follow-up. Long-term chances of seizure freedom were illustrated using a survival analysis, and predictors of recurrence were identified using Cox proportional hazard modeling.

Key Findings: Sixty-three patients with medically intractable epilepsy underwent multilobar surgical resections during the study period (mean follow-up of 4.6 years). Predominant resection types included extended occipital (temporoparietooccipital, parietooccipital, temporooccipital: 57%), frontotemporal (21%), and temporoparietal (17%). Mean age at surgery was 21.4 years and mean age at seizure onset was 10.1 years. Fifty-six percent of the patients underwent extraoperative invasive electroencephalography (EEG) evaluations. At 6 postoperative months, 71% (95% confidence interval (CI) 65–77) were

seizure-free (SF), 64% (CI 58–70) were SF at 1 year, 52% (CI 46–59) were SF at 5 years, and 41% (CI 32–50) remained SF at 10 years. Forty-one patients had at least one breakthrough seizure after surgery (median timing of recurrence 6.1 months), with an Engel class I achieved again by last follow-up in 12 of these 41 cases. Nine patients required a reoperation. Patients who underwent extended occipital/posterior quadrant resections had more favorable outcomes as compared to the other groups. With multivariate analysis, the type of resection ($p = 0.03$), preoperative auras ($p = 0.03$), an incomplete resection (0.03), and the presence of postoperative spikes ($p = 0.0003$) correlated with seizure recurrence. The risk of seizure recurrence for an incomplete resection was 2.3 (CI 1.53–3.36), preoperative aura 2.3 (CI 1.34–3.87), and postoperative spikes on surface EEG 2.5 (CI 1.29–4.71).

Significance: A favorable outcome can be achieved in 41% of patients undergoing multilobar resections for epilepsy surgery at 10 years of follow-up. Close to one-third of patients who have breakthrough seizures after surgery are able to regain seizure freedom by last follow-up. Predictors of recurrence include resection type (frontotemporal and parietotemporal resections did worse), presence of preoperative aura, an incomplete surgical resection, and the presence of postoperative interictal discharges on EEG.

KEY WORDS: Cortical dysplasia, Frontotemporal, Temporoparietooccipital, Posterior quadrant.

Resective surgery for medically intractable epilepsy has been shown to improve seizure control, quality of life, and to decrease mortality (Spencer & Huh, 2008). Although seizure outcomes following unilobar epilepsy surgery are well described, little is known about longitudinal/long-term outcomes following multilobar resections, which constitute 12–22% of large epilepsy surgery series in children (Cossu et al., 2008; Hemb et al., 2010), and 3–9% in series containing pediatric and adult patients (Eriksson et al., 1999; Rydenhag & Silander, 2001). Studies focusing solely on

these resection types have been rare, and instead, the available data are based on subsets of large resection series in the pediatric age group (Paolicchi et al., 2000; Cossu et al., 2008; Hemb et al., 2010), extratemporal epilepsy surgeries (Elsharkawy et al., 2008), and focal cortical dysplasia (Fauser et al., 2008).

Common etiologies leading to multilobar epilepsy include perinatal insults, head trauma (Koszewski et al., 1998), and type II focal cortical dysplasia (Mathern, 2009). Although these pathologies typically correlate with a high prevalence of baseline neurologic deficits, the rate of unexpected complications following multilobar resections can reach up to 4.8% when defined as leading to a neurologic deficit or affecting the activities of daily living for a period longer than 3 months (Rydenhag & Silander, 2001). As a result, a confident assessment of the risk–benefit ratio

Accepted August 11, 2011; Early View publication September 28, 2011.

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following resection of these extensive pathologic processes remains unclear.

The aim of the current study was to identify long-term seizure control in a population of patients undergoing a multilobar surgery for medically intractable epilepsy. We also aimed to identify predictors of seizure recurrence and perform a survival analysis illustrating the long-term chances of seizure-freedom.

METHODS

Patient selection

A retrospective chart review of patients undergoing surgery for medically refractory epilepsy at the Cleveland Clinic Epilepsy Center from 1994 until 2010 was performed. Only patients who underwent a multilobar resection and had at least 6 months of follow-up were included. Patients who underwent a hemispherectomy were excluded.

Data collected included demographics, neuroimaging, and preoperative and postoperative electrophysiologic variables. The occurrence of acute postoperative seizures (APOS) as defined by seizures occurring within the first postoperative week was also recorded.

Preoperative testing

Noninvasive EEG

All patients underwent a detailed history and neurologic examination. Video-electroencephalography (EEG) monitoring was performed with surface electrodes placed according to the International 10–20 system with sphenoidal electrodes used in most adults. We noted the presence of multiple spike populations, multiple ictal patterns, spikes contralateral to the side of resection, and spikes outside the lobar areas resected.

Imaging

Preoperative imaging included a high-resolution magnetic resonance imaging (MRI) 1.5 Tesla (SP system; Siemens, Erlangen, Germany) using a standardized epilepsy protocol that always included high-resolution T₁-weighted volume acquisition, T₂-weighted sequence, and fluid attenuated inversion recovery (FLAIR) sequences. Patients' MRI studies were then classified as being normal, abnormal (unilobar), abnormal (unilateral-multilobar), or abnormal (bilateral). Selected patients also underwent fluorodeoxyglucose (FDG)-positron emission tomography (PET) scan. PET scans were classified as: normal, abnormal (unilateral), and abnormal (bilateral).

Invasive EEG

Patients underwent invasive monitoring with subdural and occasionally depth electrodes to confirm the localization of the ictal onset zone because of lack of adequate information on scalp EEG and/or discordance between EEG and

neuroimaging findings. Invasive monitoring was also used to perform functional mapping preoperatively if the ictal onset zone/lesion was thought to be close to and/or overlapping with eloquent cortex. The presence of multiple spike populations, spikes outside the resection area, and contralateral spikes was noted. In addition, we documented whether the multiple spike populations or ictal areas of onset were contiguous in their location when present in different lobes on invasive EEG. We also noted whether the ictal onset zone was at the edge of the grid.

Twenty-two patients underwent intraoperative electrocorticography (ECoG) with or without electrical cortical stimulation mapping to either confirm removal of epileptogenic cortex, or to avoid resection of eloquent cortex.

Surgery

Surgical resections were classified based on the lobes involved (frontal, parietal, temporal, or occipital) and were then categorized into frontotemporal (FT), temporoparietal (TP), frontoparietal (FP), and occipital plus [temporoparietooccipital (TPO), parietooccipital (PO), or temporooccipital (TO)]. The occipital plus group represented extended posterior quadrant resections as opposed to the more anterior resection subsets.

Pathology

Pathologic specimens were examined and then divided into seven categories: Malformations of cortical development (MCDs) with abnormal MRI, MCDs with negative MRI, tumors, vascular anomalies, encephalomalacia following stroke or trauma, mesial temporal sclerosis (MTS), and cryptogenic.

Postoperative evaluation

All patients were seen at 6 weeks, 6 months (with a 2-h outpatient EEG), 1 year, and then yearly after surgery. Those with recurrent seizures were seen more frequently (every 3–6 months). In between clinic visits, patients were asked to report breakthrough seizures.

All patients had at least one postoperative MRI to determine the completeness of lesion resection. If the MRI was nonlesional, the resection was felt to be complete if it involved the areas covered by the electrodes of ictal onset and most frequent interictal abnormalities defined by subdural electrodes/intraoperative ECoG evaluations.

Definition of outcomes

Patients were classified as seizure free if they achieved an Engel class I rating at last follow-up. The primary outcome was the time to first seizure recurrence with APOS being excluded. Isolated auras were not counted as seizures. Multiple seizures occurring within 24 h were counted only once. A distinction between early recurrences defined as seizure recurrences within <6 months after surgery as opposed to late recurrences was also made.

Statistical analyses

Descriptive statistics were first obtained with means, medians, and standard deviations for continuous variables and frequencies for categorical variables. A univariate analysis was then performed using Wilcoxon rank sum, chi-square, and Fisher's exact tests to compare seizure-free patients to those with seizure recurrence, regardless of follow-up time. Variables with 10% significance on univariate analysis were then tested in a multivariate Cox proportional hazards regression model. Results were then considered statistically significant at the 5% level.

Kaplan-Meier survival analysis was first used to calculate the probability of seizure freedom in the overall group prior to any outcome predictor analysis, and later by considering each of the significant risk factors. Statistical significance was tested using the log-rank test and comparison of 95% confidence intervals (CIs).

RESULTS

Preoperative clinical characteristics

Clinical data on 63 patients who underwent a multilobar resection during the study period were reviewed and analyzed. The demographics of the cohort and follow-up duration are presented in Table 1.

Prior to surgery, the patients experienced a mean of 82 seizures per month with auras reported 40% of the time, and only two patients reporting multiple types of auras. Preoperative auras consisted of: psychic (9), visual (6), somatosensory (3), unclassified (3), abdominal (2), gustatory (2), auditory (1), and cephalic (1) sensations.

Seizure semiology consisted predominantly of simple motor (32 of 63) and automotor/dileptic seizures (30 of 63); other semiologies included complex motor (14 of 63), and hypomotor seizures (6 of 63). Multiple seizure semiologies were reported in 35% of the patients.

On examination, 19% of the patients had baseline visual deficits, and 24% had baseline hemiparesis or monoparesis.

Table 1. Demographics and seizure history

	(n = 63)
Female	29 (46%)
Mean age at seizure onset in years (range)	10.14 (0.1–45)
Mean age at surgery in years (range)	21.4 (1.3–56.0)
Mean seizure frequency in months (range)	82 (0.3–600)
Mean GTC frequency in months (range)	5 (0.1–90)
Follow-up duration in years	4.6 (0.5–14.2)
Auras present	25 (40%)
Multiple auras	2 (3%)
Generalized tonic-clonic (GTC) seizures	46 (73%)
Positive family history of epilepsy	14 (23%)
History of febrile seizures	2 (3%)
Prior epilepsy surgery	8 (13%)

Preoperative imaging and neurophysiology

Neuroimaging and electrophysiologic variables are summarized in Tables 2 and 3. In 13 patients (21%) electrocorticography was the only invasive recording and in 9 (14%) it was performed after an extraoperative invasive recording.

Invasive EEG showed multiple spike populations in 94% (32 of 34) of the cases and 55% (16 of 34) had multiple ictal patterns (Table 3). The multilobar interictal spike populations were in contiguous locations in 38% (12 of 32), and 33% (11 of 34) had spikes outside the resection area. The multilobar areas of ictal onset were in contiguous locations in 31% (5 of 16), and 21% (7 of 34) had ictal onset outside the resection area. Of the ictal patterns 29% (10 of 34) had an onset at the edge of the subdural grid.

As compared to the other resection groups, patients with occipital plus resections had less invasive monitoring as compared to the other groups ($p = 0.0075$), were more likely to have surgery before the age of 16 ($p < 0.0001$), and reported earlier onset of epilepsy ($p = 0.012$).

Type of surgery and pathology

Surgical variables are presented in Table 4. The most common etiologies for the epilepsy included MCDs in 46%

Table 2. Preoperative imaging characteristics

PET (n = 56) (%)	
Normal	6 (11)
Abnormal	50 (89)
Ipsilateral	36 (64)
Bilateral	14 (25)
MRI (n = 63) (%)	
Normal	10 (16)
Abnormal	53 (84)
Unilobar	16 (25)
Multilobar	31 (49)
Bilateral	6 (10)
Presence of MTS	4 (6)
MTS, mesial temporal sclerosis.	

Table 3. Preoperative EEG

	N = 63 (%)
Surface interictal spikes (n = 61)	
Multiple spike populations	27 (43)
Spikes outside resection area	25 (41)
Contralateral spikes	19 (30)
Surface ictal EEG	
Multiple ictal patterns	7 (11)
Use of invasive EEG electrodes	34 (56)
Use of ECoG	22 (35)
Invasive interictal EEG (n = 34)	
Multiple spike populations	32 (94)
Spikes outside resection area	13 (43)
Invasive ictal EEG (n = 29)	
Multiple ictal patterns	16 (55)
Pattern outside resection area	6 (20)
ECoG, electrocorticography.	

Table 4. Postoperative characteristics

	N = 63 (%)
Left-sided resection	35 (56)
Resection of >2 lobes	25 (40)
Type of resection	
Occipital plus (TPO, PO, TO)	36 (57)
FT	13 (21)
TP	11 (17)
FP	3 (5)
Immediate postoperative seizures	10 (16)
Complete resection	53 (84)
Contralateral spike on postop EEG (n = 54)	5 (9)
Ipsilateral spike on postop EEG (n = 54)	25 (46)

FP, frontoparietal; FT, frontotemporal; PO, parietooccipital; TO, temporooccipital; TP, temporoparietal; TPO, temporoparietooccipital.

Table 5. Engel class after 1 year and at last follow-up

	1 year (%)	Last follow-up (%)
Engel I	38 (61)	33 (52)
Engel II	4 (6)	5 (8)
Engel III	9 (14)	9 (14)
Engel IV	11 (18)	16 (25)

(29 of 63: 23 patients with abnormal MRI, six with normal MRI), encephalomalacia due to trauma or infarct in 24 of 63 (38%), tumors in 5 of 63 (8%), and others (three cryptogenic, one vascular, one mesial temporal sclerosis) in 5 of 63 (8%).

Postoperative course and outcome

Of the 63 patients included in the study, 10 (16%) developed acute postoperative seizures. A favorable outcome defined as an Engel class of I at last follow-up was achieved in 52% of these patients (Table 5).

As shown in Fig. 1, a Kaplan-Meier survival curve revealed that 71% (95% CI 65–77) had a favorable outcome at 6 months, 64% (95% CI 58–70) at 1 year, 52% (95% CI 46–59) at 5 years, and 41% (95% CI 32–50) at 10 years.

Analysis based on resection type revealed that occipital plus resections (TPO, PO, or TO) had better outcomes: 71% of these patients were seizure free 5 years after surgery as compared to only 22% and 27% seizure freedom for the frontotemporal and temporoparietal groups, respectively (Fig. 2A).

Complete seizure freedom was achieved in 33% (21 of 63) of the cases; however, even in the 42 patients who had recurrence, 12 of them could still achieve seizure control with medications alone. Patients who had seizure recurrence <6 months after surgery were less likely to achieve an Engel class I outcome at last follow-up, as only 4 of 20 patients with early seizure recurrence became seizure free at last

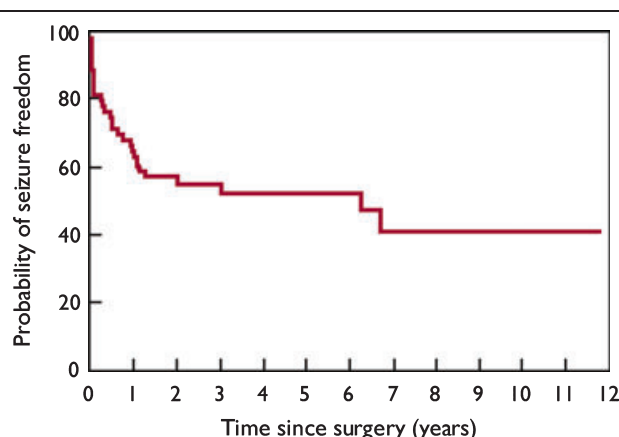


Figure 1.
Survival plot for all resection types.
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follow-up as compared to those 8 of 22 patients with later seizure recurrence who achieved seizure control at last follow-up. Overall, at last follow-up, 33 patients (52%) achieved seizure freedom.

Postoperatively, new or worsened visual field deficits, and a new or worsened motor deficit, occurred in 19% and 11% of the patients, respectively.

Nine patients from our cohort required a re-operation (mean time to operation 4.1 years), whereas another group of nine patients had been weaned from antiepileptic drugs at last follow-up. The routine EEG performed at 6 months follow-up showed ipsilateral interictal spikes in 46% of the patients (Table 4).

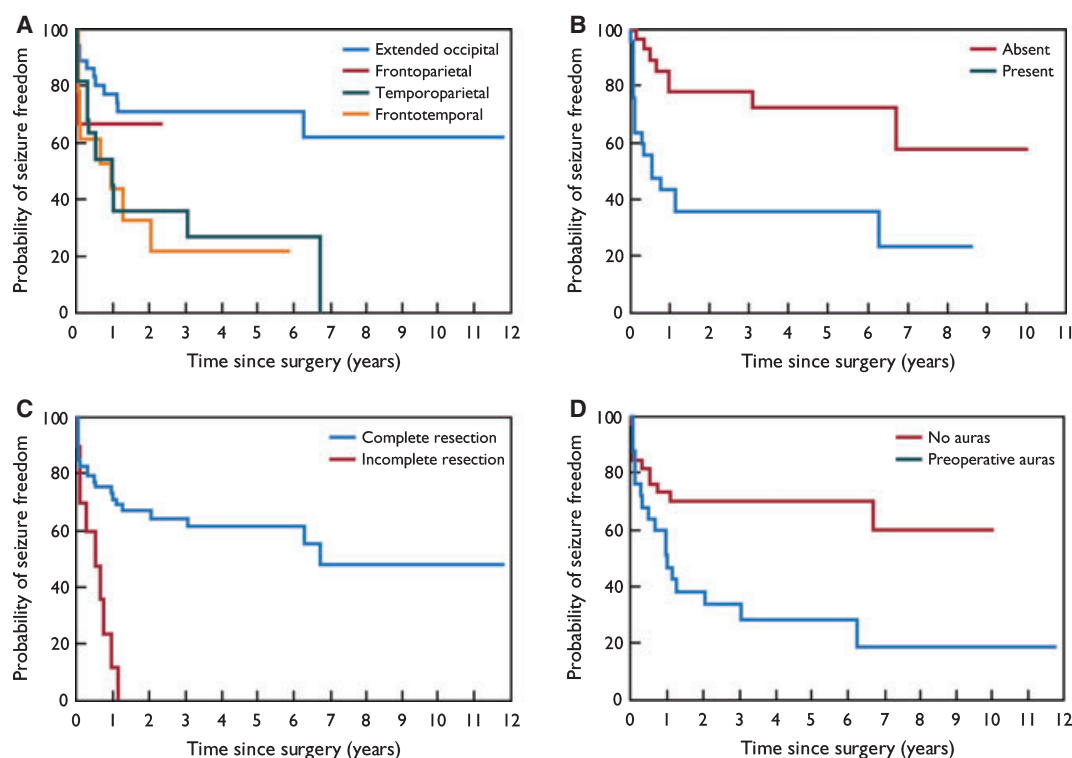
Univariate and multivariate analysis

Using univariate analysis, the parameters shown in Table 6 were found to be statistically significant predictors of seizure recurrence. With multivariate analysis, the type of resection ($p = 0.03$), preoperative auras ($p = 0.03$), incomplete resection ($p = 0.03$), and the presence of postoperative spikes ($p = 0.0003$) correlated with seizure recurrence. The risk of seizure recurrence for an incomplete resection was 2.3 (95% CI 1.53–3.36), preoperative aura 2.3 (95% CI 1.34–3.87), and postoperative ipsilateral spikes on surface EEG 2.5 (95% CI 1.29–4.71). Longitudinal outcomes based on these factors are illustrated in Fig. 2A–D.

DISCUSSION

Seizure-free outcomes

This is the first study reporting long-term longitudinal outcomes in patients with multilobar epilepsy surgery. Following multilobar resections in a group of adult and pediatric patients with medically intractable epilepsy due to various types of pathologies, seizure freedom was achieved in 41% at a follow-up of 10 years. In prior studies, seizure

**Figure 2.**

(A–D) Survival plot based on surgery type (A), postoperative ipsilateral spike on EEG (B), completeness of resection (C), and presence of preoperative auras (D).

Epilepsia © ILAE

Table 6. Predictors of seizure outcome (univariate analysis)

	Seizure free (n = 33) (%)	Recurred (n = 30) (%)	p-value
Age of seizure onset <16 years (n = 48)	20 (42)	28 (58)	0.09 ^a
Presence of auras (n = 25)	7 (28)	18 (72)	0.0014 ^a
Prior epilepsy surgery (n = 8)	6 (75)	2 (25)	0.09 ^a
Type of resection			
Occipital plus (TPO, PO, TO) (n = 36)	25 (69)	11 (31)	0.0072 ^a
FT (n = 13)	4 (31)	9 (69)	
TP (n = 11)	2 (18)	9 (82)	
FP (n = 3)	2 (66)	1 (33)	
Complete resection (n = 53)	32 (60)	21 (40)	0.002 ^a
Spikes on postoperative EEG (n = 25)	8 (32)	17 (68)	0.0026 ^a

FP, frontoparietal; FT, frontotemporal; PO, parietooccipital; TO, temporooccipital; TP, temporoparietal; TPO, temporoparietooccipital.

^ap < 0.10.

free outcomes at last follow-up ranged between 22% and 55% (Patil et al., 1997; Eriksson et al., 1999; Paolicchi et al., 2000; Elsharkawy et al., 2008; Hemb et al., 2010). The seizure-free rate of 41% at 10 years noted in this study

is comparable to reports of 44.4% in series of extratemporal multilobar resections with a similar follow-up period (Elsharkawy et al., 2008). The discrepancy with the other studies may be due to surgical technique; Patil et al. (1997) focused on multiple subpial transections plus topectomy, whereas others reported on the outcome in different patient populations; age <12 in Paolicchi et al. (2000) and strictly pediatric patients in Hemb et al. (2010).

The lack of longitudinal follow-up in many of the prior studies may lead to more optimistic conclusions about surgical outcome when in fact the results may not necessarily be sustained over time. As our data showed, approximately half of the surgical failures after multilobar resections occurred during the first 6 months after surgery and the other half of the failures happened between 6 months and 10 years after resections.

Other studies have reported higher seizure-free rates with unilobar resections in different patient populations (Paolicchi et al., 2000; Fauser et al., 2008; Hemb et al., 2010).

Compared to unilobar surgeries at our center, seizure freedom in 52% of the patients at 5 years was worse than that observed in patients following either temporal lobe (63%) or posterior quadrant (63%) resections (Jeha et al., 2006; Jehi et al., 2009). However, it was better than that

following frontal lobe resection (30% at 5 years) in patients in whom medically intractable frontal lobe epilepsy was suspected (Jeha et al., 2007). Clearer trends, however, emerge when the types of resections are analyzed. Extended occipital resections achieved 71% seizure freedom at 5 years, which is even better than in their unilobar counterparts. On the other hand, frontotemporal resections achieved a seizure freedom identical to that observed after unilobar frontal resections.

Although 52% of our patients eventually achieved a favorable outcome, it was interesting to note that 67% of the patients had at least one seizure postoperatively. Seizure recurrence occurred in 48% of the patients during the initial 6 month postoperative period. Patients with late recurrence had a higher chance of achieving seizure freedom as compared to those with early postoperative seizure recurrence. This is similar to our finding in unilobar patients, where the occurrence of two seizures within the 6-month postoperative period was associated with refractoriness (Jeha et al., 2010). Although the reasons behind these differences between the early and late seizure recurrence on the likelihood of seizure outcome at last follow-up are not clear, we suspect that the early recurrence of seizures would probably be due to incomplete resection of the epileptic focus. The late seizure recurrence in this patient population raises the question of epilepsy progression (Sperling et al., 2008). With the limitations of current technology, the seizure-onset zone can be well defined and resected (Rosenow & Luders, 2001). However, no methods currently exist to delineate the “potential seizure onset zone,” which may be the source of this delayed recurrence.

Predictors of recurrence

Patients with an extended occipital resection (occipital plus) had the best seizure-free rates at last follow-up. Our results of an Engel class I of almost 70% at 5 years and 60% at 10 years are similar to those reported Binder et al. (2008), where 70% of patients with extended occipital resections had seizure freedom with a mean follow-up of 6.7 years.

These favorable results contrasted with the extended temporal resections where frontotemporal resections achieved 22% seizure freedom and with temporoparietal resections with 27% seizure freedom at 5-year follow up.

A more aggressive TPO resection will have a higher likelihood of removing the epileptogenic zone. The patients in our cohort had earlier seizure onset and earlier resections as compared to the other groups. The presurgical hypothesis was also clearer, and as a result less invasive monitoring was needed.

Furthermore, epilepsies arising from the more posterior areas likely have better localization due to the absence (or limited presence) of spread patterns to those areas from other regions of the brain.

Occipitotemporal pathways are either direct through the inferior longitudinal fasciculus or indirect through polysynaptic U fibers (Catani et al., 2003). A number of studies have described ictal spread patterns from the occipital lobe with only rare reports of spread patterns to the occipital lobe (Jacobs et al., 2008).

In their cohort of 25 patients with occipital lobe epilepsy, Williamson et al. (1992) were able to document occipitofrontal spread and occipitotemporal spread in 3 and 12 patients, respectively.

In their analysis of 112 seizures of different areas of origin, only one seizure, which was of parietal onset, spread to the occipital lobe (Jenssen et al., 2011).

On the other hand, the more anterior regions have richer connections, and spread patterns to temporal areas have been described from cingulate (Koubeissi et al., 2009), insular (Isnard et al., 2004), and orbitofrontal cortices (Shihabuddin et al., 2001). Insular epilepsy can also falsely localize to the frontal or temporal lobes (Levitt et al., 2010). The possibility remains that with patients who are receiving the indication for resections extending outside the temporal lobe, a more widespread network that includes deeper structures such as the insula is missed (Ryvlin & Kahane, 2005).

Extended frontotemporal resections (after failed frontal lobe surgery) have also been shown previously to be ineffective in achieving seizure freedom (0 of 13) (Salanova et al., 1994).

A complete surgical resection of the epileptogenic tissue, whether based on imaging or intraoperative ECoG, was also associated with better rates of seizure freedom. This has previously been shown in a large series of patients with focal cortical dysplasia (Krsek et al., 2010), and is consistent with the findings in our cohort of frontal lobe epilepsy surgeries (Jeha et al., 2007). In contrast, surgical approaches attempting to spare lesional eloquent cortex run the risk of worse outcomes (Sarkis et al., 2010).

The presence of preoperative auras was a predictor of recurrence in our series. This may have been due to their false-localizing influences. It is possible that the temporal symptomatogenic zone may have been the result of spread from a silent ictal-onset zone and subsequently may have led to the generation of an inaccurate epileptogenic zone hypothesis and surgical exploration. Elsharkawy et al. (2008) noted that the presence of preoperative auditory auras was a predictor of recurrence in their extratemporal surgery series. Almost half of the auras reported in our patient cohort were suggestive of possible temporal lobe onset. Because these auras may be at times mislocalizing of the seizure onset, this may explain the higher rates of seizure recurrences in patients who underwent frontotemporal and temporoparietal resections.

Interictal spikes on postoperative EEG have been established as a useful predictor of seizure recurrence in both temporal and extratemporal epilepsy surgery (Jeha et al.,

2010; Rathore & Radhakrishnan, 2010). Our results extend its usefulness to patients with multilobar surgeries as a marker of residual epileptogenic tissue.

CONCLUSION

A long-term (10 years) favorable outcome can be achieved in 41% of adult and pediatric patients with severe and medically intractable epilepsy and who underwent multilobar surgical resections. Even in patients with recurrence after surgery (especially in those with recurrences later than 6 months after surgery), 29% can still achieve an Engel class I outcome with medications alone.

Predictors of seizure recurrence following multilobar surgeries included frontotemporal and parietotemporal resections, presence of preoperative aura, an incomplete surgical resection, and postoperative interictal discharges.

DISCLOSURE

None of the authors has any conflict of interest to disclose. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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